Tutorial Sheet No. 12

Electricity

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Abstract

This tutorial strengthens conceptual mastery of **electric fields**, **current electricity**, and **DC circuits**. Thinking questions are designed to highlight physical reasoning, common misconceptions, and the meaning behind key formulae used in quantitative problems.

1 Electric Fields

Theory – Electric Field and Potential

The electric field **E** at a point is defined as the force experienced per unit positive charge:

$$\mathbf{E} = \frac{\mathbf{F}}{q}.$$

Electric potential V is the potential energy per unit charge. A charge naturally moves from high potential to low potential.

a) **Direction Misconception** A student says: "A negative charge moves from high potential to low potential." Explain why this statement is incorrect, and state the correct behaviour.

b) **Field Line Reasoning** Draw (in words) the electric field pattern around a positive and a negative charge separated by some distance. Explain why field lines never cross.

c) **Strength of Field** Two points A and B are in a uniform electric field. The potential difference $V_A - V_B$ is the same regardless of the path taken. Explain why this is true for electric fields but not necessarily for other force fields.

d) **Charged Droplet Thought Experiment** A small oil droplet carrying charge $-q$ is suspended motionless between two horizontal plates. Describe the force balance involved and what happens if the droplet becomes *positively* charged instead.

2 Current Electricity

Theory – Current, Resistance, and Energy

Electric current is the rate of flow of charge:

$$I = \frac{Q}{t}.$$

The heating effect of current depends on power:

$$P = VI = I^2R = \frac{V^2}{R}.$$

a) **Resistance and Temperature** Give one reason why resistance increases with temperature for a metal but decreases for a thermistor.

b) **Unit Confusion Clarification** A student says: "Voltage is the amount of charge that passes a point per second." Explain the mistake and give the correct interpretation of voltage.

c) **Energy Reasoning** Two bulbs have the same resistance. One is connected to a 3 V supply, the other to a 6 V supply. Without calculations, compare their brightness and justify using energy ideas.

3 DC Circuits

Theory – Series and Parallel Circuits

In series circuits, current is the same everywhere. In parallel circuits, voltage across each branch is equal.

a) **Series vs. Parallel – Subtle Case** Two identical bulbs are first connected in series, then in parallel. Explain why bulbs in parallel glow more brightly even though their resistance does not change.

b) **Internal Resistance Insight** If the internal resistance of a battery increases, explain why the brightness of bulbs connected to it decreases.

c) **Potential Divider Reasoning** A potential divider has two resistors R_1 and R_2 in series. A student claims: "If R_1 increases, the output voltage across R_2 must also increase." Determine whether this statement is correct and explain why.

d) **Short Circuit Danger** Describe what happens to the current, the battery, and the wires when a circuit is accidentally shorted. Explain why this is dangerous using resistance and power arguments.

4 Challenge Question – Linking Concepts

Extension: Fields, Energy, and Circuit Behaviour

Consider a charged particle moving through a region with both an electric field and a conducting path.

a) A particle starts from rest at a point of high electric potential. Explain why it must gain kinetic energy as it moves, and what determines the final energy gained.

b) Now suppose the particle travels through a wire. Explain why its drift velocity becomes very small even though the electric field can be large.

c) Discuss how the energy lost by the particle in the wire ultimately appears as heat, linking the ideas of potential difference, work done, and power dissipation.

Formula Cheat Sheet – Quick Revision

Summary: Key Equations and Relations

Electric Fields

$$E = \frac{F}{q}, \qquad V = \frac{W}{q}, \qquad E = \frac{V}{d}$$

Current Electricity

$$I = \frac{Q}{t}, \qquad V = IR, \qquad P = VI = I^2R = \frac{V^2}{R}$$

DC Circuits

$$R_{\text{series}} = R_1 + R_2, \qquad \frac{1}{R_{\text{parallel}}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$V_{\text{out}} = \frac{R_2}{R_1 + R_2} V_{\text{in}}$$